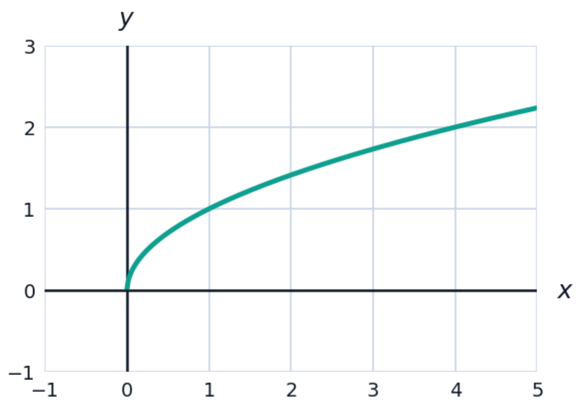
Area Between Curves

Area between two curves

1 Find the area of the region bounded by $y = x^2$ and $y = x + 2$.

2 The region bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$ is shown. Find its area.



Setting up with respect to y, and total area

3 Find the area enclosed by $x = y^2$ and $x = 2y$ by integrating with respect to y.

4 Find the total area between $f(x) = \sin x$ and the x-axis on $[0, 2\pi]$ (note the curve is below the axis on part of this interval).

More area problems

5 Find the area of the region bounded by $y = x^2 - 4$ and $y = 4 - x^2$.

6 Find the area between $y = x^3$ and $y = x$ on $[0, 1]$ (note that $y = x$ is above $y = x^3$ on this interval).

Areas requiring case splits

7 Find the area between $y = x^3$ and $y = x$ on $[-1, 1]$ (note the curves cross at $x = 0$, so the identity of the upper curve switches).

8 Find the area enclosed between $y = 2x$ and $y = x^2$ on the interval where $y = 2x$ is above $y = x^2$.

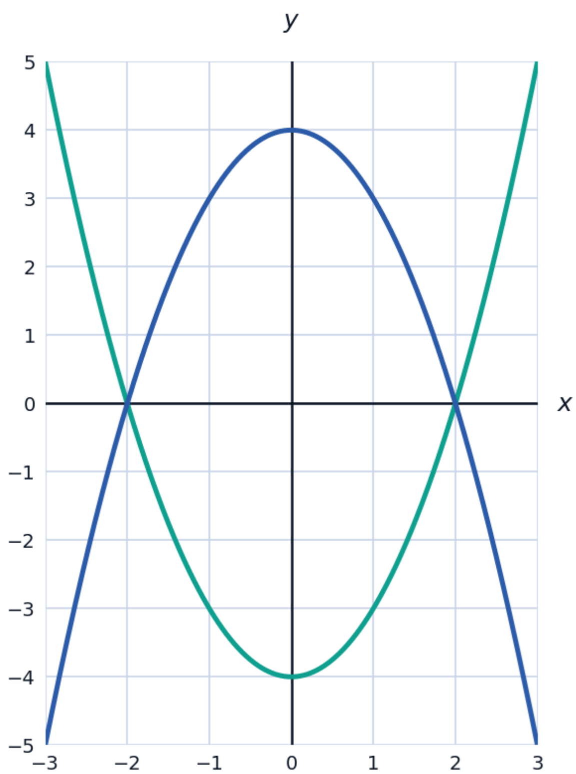
Areas involving trigonometric and exponential curves

9 Find the area between $y = \cos x$ and $y = 0$ on $\left[0, \frac{\pi}{2}\right]$.

10 Find the area between $y = e^{x}$ and $y = e^{-x}$ on $[0, 1]$ (note $e^{x} \geq e^{-x}$ on this interval).

Visualizing a region bounded by two curves

11 The graphs of $f(x) = x^2 - 4$ and $g(x) = 4 - x^2$ from an earlier problem are shown together. Confirm from the picture that the region is symmetric about the y-axis, and use this to verify the earlier area answer $\frac{64}{3}$ using $2\int_0^2 (8 - 2x^2)\,dx$.



Area between a curve and a horizontal line

12 Find the area between $y = 6 - x^2$ and $y = 2$.

13 Find the area between $y = \sqrt{x}$ and $y = \frac{x}{2}$ for $x \geq 0$.

Area between curves in applied contexts

14 Two runners' velocities (m/s) over a 10-second race are $v_1(t) = 2t$ and $v_2(t) = t + 3$. The area between these curves represents the gap in distance traveled. Find this gap over $[0, 10]$.

15 Find the area of the region bounded by $y = x^2 - 2x$ and the x-axis on $[0, 2]$ (note the curve dips below the axis on this interval).

16 Find the area between $y = x^3 - x$ and the x-axis on $[-1, 1]$, splitting at the roots $x = -1, 0, 1$ where the sign changes.